Huawei Access Node HEB-ACC-40 QoS / Congestion Event (Microbursts, interface tail drops, QoS queue starvation) Resolution SOP

Node: HEB-ACC-40 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_7886

Description

The described failure domain is a result of the complex interactions between the Huawei Access Node's QoS (Quality of Service) and congestion control mechanisms. Specifically, on this Node, microbursts and interface tail drops may occur due to the misconfigured QoS policies, resulting in priority queue starvation. This phenomenon is further exacerbated by the Huawei Access Node's reliance on the Weighted Random Early Detection (WRED) algorithm for congestion avoidance, which may not adequately handle bursty traffic. Additionally, the Huawei Access Node's architecture, which employs a hierarchical QoS model, may lead to unintended consequences when multiple QoS policies interact. This can result in suboptimal traffic handling, ultimately manifesting as QoS queue starvation and congestion events.

Pre-requisite Checks

- Verify the Node's physical and environmental status, including temperature, humidity, and power supply.

- Confirm the redundant link status to ensure the Node's QoS policies are not inadvertently impacting traffic distribution.

- Check the Huawei Access Node's software version and ensure it is up-to-date.

- Verify the Node's QoS policies are correctly configured and not causing any anomalies.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
WRED algorithm misconfiguration	Medium
Hierarchical QoS model misconfiguration	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos policy, Expected Output: QoS policy configuration and statistics, Description: This command will display the current QoS policy configuration and statistics, allowing for the identification of potential misconfigurations or issues.

Command: display interface qos, Expected Output: Interface QoS configuration and statistics, Description: This command will display
```

```
the current QoS configuration and statistics for each interface, aiding in the identification of potential issues related to interface QoS.

Command: display wred policy, Expected Output: WRED policy configuration and statistics, Description: This command will display the current WRED policy configuration and statistics, enabling the identification of potential misconfigurations or issues related to the WRED algorithm.
```

Resolution Steps

Step 1: Step: Review and adjust QoS policies, Command: undo qos policy , Description: Review the QoS policies and adjust them as necessary to prevent misconfigurations or issues. This may involve reconfiguring policy priorities, bandwidth allocations, or other settings.

Step 2: Step: Reconfigure WRED policy, Command: undo wred policy , Description: Review the WRED policy and reconfigure it as necessary to prevent misconfigurations or issues. This may involve adjusting the WRED weights, thresholds, or other settings.

Step 3: Step: Verify hierarchical QoS model configuration, Command: display qos policy hierarchy, Description: Verify that the hierarchical QoS model is correctly configured and not causing any unintended consequences.

Post-Check Verification

```
Command: display qos policy, Expected Output: Verified QoS policy configuration and statistics, Description: Verify that the QoS policies are correctly configured and not causing any issues.

Command: display interface qos, Expected Output: Verified interface QoS configuration and statistics, Description: Verify that the interface QoS configuration is correct and not causing any issues.

Command: display wred policy, Expected Output: Verified WRED policy configuration and statistics, Description: Verify that the WRED policy is correctly configured and not causing any issues.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: undo undo qos policy , Description: If the fix fails, restore the original QoS policy configuration.

- Step: Restore original WRED policy configuration, Command: undo undo wred policy , Description: If the fix fails, restore the original WRED policy configuration.